

Online Neural Space Time Memory for Dynamic Novel View Synthesis

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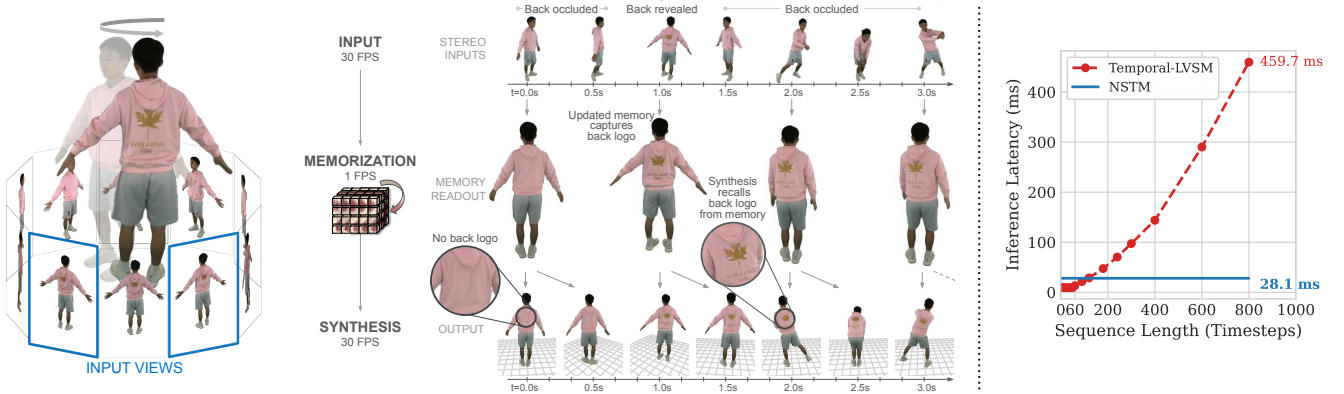


Figure 1. **Neural Space-Time Memory (NSTM)** synthesizes novel views in real-time from 30 FPS multi-view videos, while updating its memory at 1 FPS. **Middle:** NSTM captures the back logo as the subject rotates (see memory readout), faithfully recalling it later from memory given only frontal views. **Right:** While full self-attention in typical feed-forward models (e.g., LVSM [22]) scales quadratically with history sequence length, NSTM achieves real-time inference with $\mathcal{O}(1)$ complexity at 256×256 resolution on one H100 GPU.

Abstract

Online novel view synthesis from multi-view streaming videos faces a fundamental trade-off: maintaining a persistent, long-horizon memory to reconstruct temporarily occluded regions while operating under strict real-time constraints. While Test-Time Training (TTT) offers a powerful memory mechanism, standard models mandate gradient-based memory updates at every frame to adapt to the changing motion in dynamic scenes. The computational cost of heavy memory updates precludes real-time application and can lead to instability over long contexts. Given that memory updates are more demanding than memory application and video content is largely redundant, we propose to decouple the frequencies of these two processes. Our approach performs periodic memory updates while applying the memory on a per-frame basis, using cross-view attention to manage deformations between the prior memory state and the current frame. To lock in the historical context, we introduce two critical mechanisms: an auxiliary Memory Loss that forces persistent internalization of the scene, and a Memory Caching strategy that regularizes active weights against catastrophic drift. Our method demonstrates real-

time, state-of-the-art performance on scenes with dynamic human motion as well as minute-scale online memorization.

1. Introduction

Online novel view synthesis (NVS) of dynamic scenes from multi-view streaming videos is a highly sought-after goal in computer vision, underpinning many applications such as 3D telepresence and live free-viewpoint broadcasts. However, sparse camera setups and occlusions often make observations incomplete at a single instant. For example, synthesizing a person’s back when current inputs provide only frontal views is challenging without historical context (Fig. 1). Solving this problem requires a *space-time memory* – akin to how humans integrate information across space and time, naturally recalling previously observed, now-occluded visual details. However, current online dynamic NVS methods struggle to maintain a persistent memory over a long time without sacrificing real-time speed.

In this paper, we present an online dynamic NVS system from multi-view videos capable of storing persistent, minute-scale historical context to perform grounded synthesis of momentarily occluded regions, all while operating under amortized real-time constraints. At the core of

our approach is a *space-time memory* mechanism based on *Test-Time Training (TTT)*. We leverage TTT for its expressive memory capacity and, crucially, its linear scalability – bypassing the prohibitive quadratic computational cost of full self-attention heavily used in typical feed-forward NVS methods (Fig. 1). Nevertheless, while TTT has shown promise in static 3D environments, adapting it to continuous, streaming dynamic scenes in real time poses several challenges, as follows.

Speed. Real-time dynamic NVS via TTT faces an architectural bottleneck. Standard TTT gains efficiency through *large-chunk* processing and per-image attention [64, 66]. However, this per-image attention restricts target views to rely entirely on the memory module for input context. Thus tracking continuous motion requires breaking efficient large chunks into per-timestep (*small-chunk*) updates. This degrades efficiency: a full per-timestep TTT step (memory update and apply) takes **58ms** at 256×256 on an H100, precluding real-time deployment. Conversely, pure memory apply requires only **27ms** (35 FPS). Exploiting this computational asymmetry and video redundancy, we propose to *decouple* their frequencies (Fig. 1): updating memory (*memorization*) periodically while applying it (*synthesis*) per frame. To resolve the resulting motion misalignment between the periodically updated memory and the current frame, we introduce a *cross-view attention* mechanism (Fig. 2) – it allows target views to directly attend to current inputs, fusing ongoing motion with past memory context to achieve amortized real-time speed.

Memory Persistence and Supervision Balance. Decoupling synthesis and memorization frequencies requires careful design of the training objective. This is because occlusion signals in natural videos are spatially and temporally sparse. Jointly training the separate memorization and synthesis tasks on this sparse data distribution results in an underconstrained problem, causing the network to over-rely on processing current, incomplete observation, effectively bypassing the memory module and forgetting the past (Sec. 4.3). To resolve this, we disentangle the training objectives and propose an auxiliary *memory loss*. During the *Memorization Step*, we query the memory using only the target view camera rays (memory tokens in Fig. 2), deliberately preventing any attention over the current input views. By isolating the readout, we force the memory module to reconstruct the scene entirely from its internalized historical context. By explicitly supervising the memory’s independent reconstruction capabilities, we force the memory module to persistently internalize the scene context.

Long-Term Stability. Continuous memory updates in TTT can lead to instability over long time horizons. Our periodic update scheme natively mitigates this by reducing the overall update frequency. To further enhance stability, we adopt a *memory caching* strategy [6] that aggregates histor-

ical memory checkpoints, stabilizing the parameter space to handle minute-long streams. Crucially, modeling deformation in the Synthesis Step enables memory caching to handle changing motion (Sec. 4.3). Finally, we identify that the standard dot-product inner loss used in state-of-the-art TTT works [23, 64, 66] causes magnitude inflation over many update steps and show that we can stabilize the framework by replacing it with an L_2 objective [3, 7].

We evaluate our framework on the large-scale MVHumanNet++ dataset [55], as human motion represents one of the most prominent and challenging applications of online dynamic NVS from multi-view videos. Despite the dataset’s diverse and complex motion and garments, NSTM successfully delivers high-fidelity synthesis and stable, minute-long persistent memorization at amortized real-time speeds.

Our key contributions are summarized as follows:

- **Real-Time, Long-Context Dynamic NVS:** We propose Neural Space-Time Memory (NSTM), the first online framework for dynamic novel view synthesis from multi-view videos that sustains minute-long persistent memory while operating in amortized real-time.
- **Decoupled Memorization and Deformation:** We overcome the TTT computational bottleneck by structurally decoupling the frequencies between periodic, heavy memory updates and continuous, transformer-driven per-frame synthesis.
- **Disentangled Memory Supervision:** We introduce an alternating *memorization* and *synthesis* training regime and an auxiliary **memory loss** that explicitly forces the network to internalize persistent scene context, overcoming sparse supervision occlusion signals.
- **Memory Caching & Stability:** We enable stable, minute-long view synthesis by aggregating historical states via **memory caching**.

2. Related Works

Novel View Synthesis. Classical NVS enforces multi-view geometric consistency through explicit 3D representations such as meshes [18, 36, 44], multi-plane images [8, 12–14], neural radiance fields [31, 34, 47, 48, 51, 61, 62], point clouds [1, 46, 50, 56], and 3D Gaussian splatting [20, 27, 54, 58]. While effective, these representations impose implicit inductive biases that constrain flexibility on scenes that deviate from their predefined priors, motivating more general data-driven architectures [37, 40]. Diffusion-based methods relax these constraints by leveraging strong 2D generative priors [19] to hallucinate unobserved regions [32, 33, 39, 53, 63], but rely on iterative sampling that is ill-suited to real-time streaming. A separate, representation-free line discards the explicit 3D proxy entirely: LVSM [22] and LaCT [66] cast NVS as a latent sequence modeling task, tokenizing source views

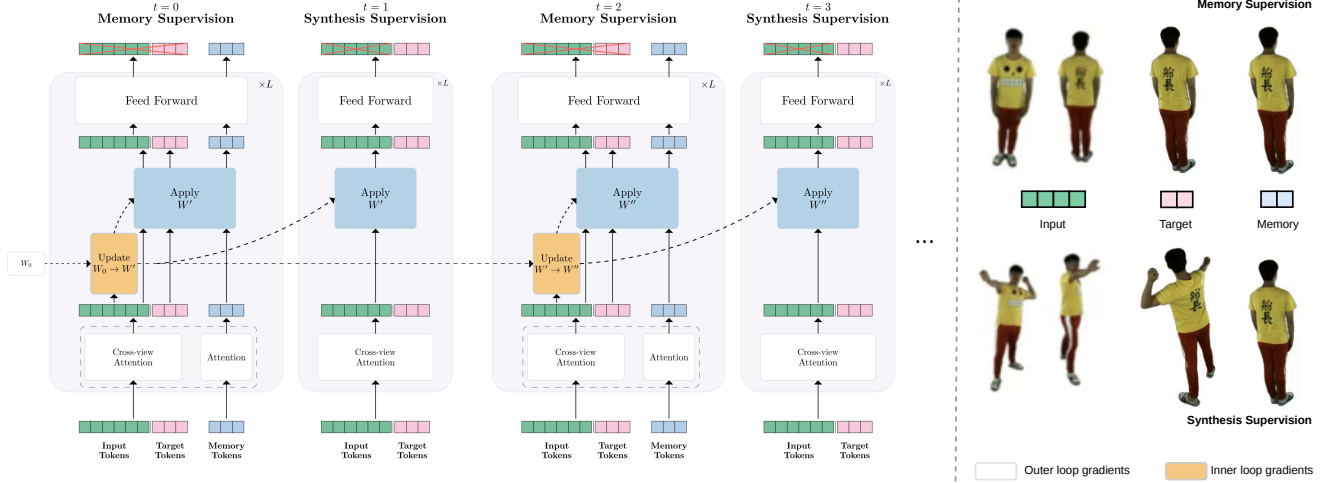


Figure 2. **NSTM Training Scheme.** **Left:** Our alternating training scheme explicitly enforces memorization. In the *Memory Supervision* step, isolated *memory tokens* (target camera rays) perform strict self-attention — sharing weights with the *cross-view attention* layer — without attending to input views. Once the memory is updated by the inputs, these isolated tokens query the memory to reconstruct the target view. This forces a pure memory readout, ensuring context internalization. During *Synthesis Supervision*, we perform self-attention across input and target tokens to capture ongoing motion, and then query the previously updated memory state. **Right:** Visualizations of the inputs, targets, and memory in both steps. Crucially, the historical memory state often exhibits a different pose from the current observation during synthesis. Our *cross-view attention* addresses this deformation, successfully fusing past appearance with current motion.

and letting expressive transformers implicitly capture multi-view consistency in a single feed-forward pass. We build on this representation-free formulation for its scalability and freedom from geometric priors. Crucially, whereas prior works target offline batch-processing of static scenes, NSTM elevates this paradigm to real-time dynamic streaming via a persistent space-time memory and our decoupled memorization-synthesis architecture.

Streaming Scene Reconstruction. Streaming novel view synthesis in a dynamic setting requires causal, efficient processing of continuous video streams. Unlike offline methods that retrospectively optimize over an entire sequence [9, 15, 29, 30, 54, 57, 60], streaming reconstruction performs online updates using only current and previous frames under strict computational budgets. Prior per-frame optimization approaches [41, 59] are too slow for real-time use, while recent work [52] accelerates this using optical flow to reuse prior information, though this local tracking accumulates errors that degrade quality over time. We address this by encoding a persistent memory directly into the network’s fast weights [66], allowing our model to retain context and memorize scene geometry for temporally consistent synthesis without excessive drift.

Memory Modules. The quadratic cost of Transformers [45] on long-context data has driven a broad search for sub-quadratic memory. Linear attention replaces the Softmax operator with a separable kernel [26], compressing context into a fixed-size matrix-valued memory, and a large body of work has since refined the *discipline* of how this memory is written and maintained: through stronger kernels [2, 25], more expressive learning rules such as the Delta [38], Oja [21], and Omega [3] rules, principled forget-

ting and retention gates [7, 17, 35, 42], and hybrid attention designs [5, 11]. Because a fixed-size matrix is capacity-limited and underperforms on retrieval, a further line based on Test-Time Memorization/Training [7, 43] and Nested Learning [4] introduces deep memories whose capacity grows with depth [3, 5], and sub-linearly growing memories via structured matrices or memory caching [6, 16]. We build directly on this memory-discipline lineage. Concretely, our memory differs from all of the above in four respects: (1) prior modules are 1-dimensional and sequential, and cannot exploit the rich space-time dependencies in multi-view videos; (2) their coupled update-apply mechanism causes suboptimal information flow along the time dimension, as our ablations confirm (Sec. 4.3, Tab. 2); (3) contrary to these approaches, our design is based on decoupling memorization and synthesis by using different modules in our architecture; and (4) we adopt an L_2 objective for long-horizon stability; this avoids the magnitude inflation that the dot-product inner loss used across most prior memory modules, including the large-chunk TTT of LaCT [66], incurs over long rollouts.

3. Method

We address online novel view synthesis from multi-view dynamic video streams. At each timestep t , we observe N input views $\mathcal{O}_t = \{(\mathbf{I}_{t,i}, \mathbf{P}_{t,i})\}_{i=1}^N$, where $\mathbf{I}_{t,i} \in \mathbb{R}^{H \times W \times 3}$ is an RGB image and $\mathbf{P}_{t,i} \in SE(3)$ is the corresponding camera pose. Because $\mathcal{O}_t$ provides an incomplete observation of the true dynamic scene (e.g., due to occlusion), NSTM synthesizes the target image $\mathbf{I}_t^{\text{tgt}}$ by conditioning the rendering function on both the current input $\mathcal{O}_t$ and an ac-

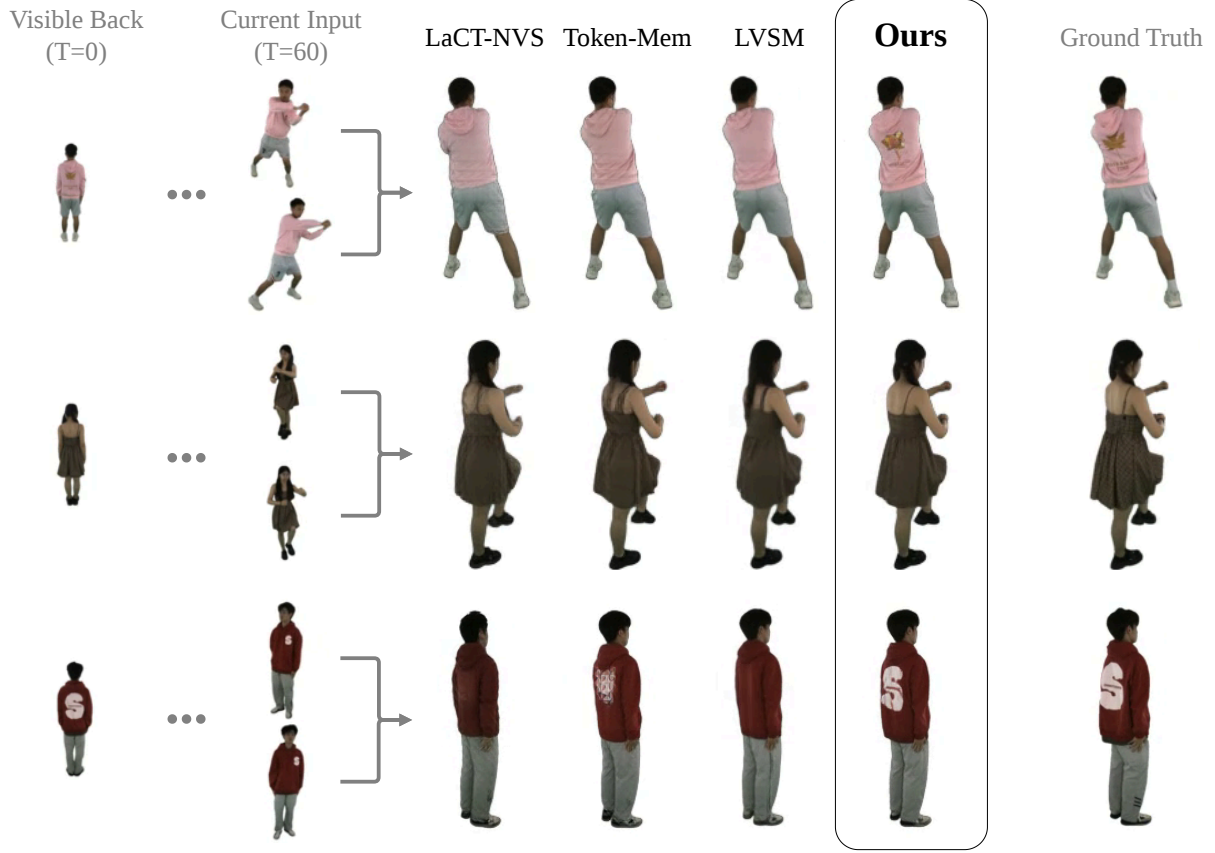


Figure 3. **Memory Stress Test at $T = 60$.** Despite only observing the back at $T = 0$, NSTM accurately recalls the hoodie patterns and hairstyles while fusing with current motion. LVSM makes its best stateless guess in unseen regions, LaCT-NVS suffers from out-of-distribution memory drift, and Token-Mem fails to recall accurate back views. See our supplementary website for the video results.

cumulated implicit memory $\mathcal{W}_{t-1}$:

$$\mathbf{I}_t^{\text{tgt}} = \Phi(\mathbf{P}_t^{\text{tgt}}, \mathcal{O}_t; \mathcal{W}_{t-1}). \quad (1)$$

Following LVSM [22], we channel-wise concatenate input RGB images with their corresponding per-pixel Plücker ray maps, using linear layers to patchify them into input tokens. Similarly, we patchify the Plücker rays of the target camera to form target tokens. These concatenated input and target tokens are then passed through a stack of Transformer blocks, each comprising an attention module, a TTT memory module, and a feed-forward MLP module (Fig. 2). Finally, we extract the processed target tokens and unpatchify them, applying a sigmoid activation to predict the target RGB images (see Sec. D for more details).

TTT memory module. Test Time Training/Memorization [7, 43] defines implicit memory as a set of *fast weights* that are updated at inference time, forming a compressive neural memory. This contrasts with the *slow weights* (i.e., model parameters) that are frozen during inference. In our architecture, we use $\mathcal{W}_{t-1} = \{W_1, W_2, W_3\}$ as the fast weights with the design of a bias-free SwiGLU-MLP:

$$f_{\mathcal{W}}(x) = (\text{SiLU}(W_1 x) \odot (W_3 x)) W_2, \quad (2)$$

Input tokens are projected into keys k , queries q , and values v , then to read from the weights we **apply** them via ($\circ$ =

$f_{\mathcal{W}}(q)$). The **update** operation memorizes context via a single gradient step on an inner loss binding keys to values,

$$\mathcal{W}' = \mathcal{W} - \eta \nabla_{\mathcal{W}} \mathcal{L}_{\text{inner}}(f_{\mathcal{W}}(k), v). \quad (3)$$

The inner loss is a design choice that distinguishes different existing architectures and determines the memory management in prioritizing the input tokens. In this paper, we use an L_2 inner loss:

$$\mathcal{L}_{\text{inner}} = \|v - f_{\mathcal{W}}(k)\|_2^2. \quad (4)$$

In Tab. 2 and Fig. 5 we show that contrary to the choice of unbounded dot-product loss as in some previous models (such as LaCT [66] or DLA [3]) that cause severe magnitude inflation of $f_{\mathcal{W}}(k)$ after repeated gradient descent over long horizons, our choice of L_2 objective results in long-horizon stability.

We further stabilize the update by Newton–Schulz orthonormalization of the gradient (Muon [24]) and L_2 weight normalization. Our network is trained end-to-end via an outer-loop task loss over the slow weights, initial fast-weight state, and projections. We denote the fast-weight state at timestep t as $\mathcal{W}_t$ (Sec. 3.3).

3.1. Decoupled Memorization and Synthesis

Naïvely applying our memory architecture summarized in Sec. 3 to live video streams introduces a latency bottleneck. Standard TTT-based model rigidly couples the memory update and apply steps. To achieve high throughput, prior static frameworks [66] process sequence chunks in parallel. However, this relies on *chunk-level buffering*, which introduces unacceptable delays and violates the strict online constraints of live streaming. Conversely, processing the stream sequentially, by forcing a heavy gradient-based memorization at every incoming frame, incurs prohibitive computational overhead of 58ms (see Tab. 4 latency breakdown), severely precluding real-time performance. To overcome this cost, NSTM decouples the frequency of memory updates from novel view synthesis, and decomposes the online inference process into two frequency-asymmetric steps (Fig. 2):

Periodic Memorization Step: Rather than updating at every frame, we strictly limit heavy gradient updates to a periodic schedule to compress novel states into the fast-weights. **Per-frame Synthesis Step:** At every timestep t , we execute a lightweight query against the persistent memory $\mathcal{W}_{t-1}$ Eq. (1). However, because the scene is dynamic and memory updates are periodic, the memory state and the current observation are inherently misaligned (Fig. 2). To handle this temporal deformation, we adopt a *cross-view attention* mechanism. Jointly attending across the source posed images and the target camera rays explicitly captures ongoing motion. The output then queries the memory module, dynamically fusing current motion with historical context to synthesize the novel view at amortized real-time speeds (Fig. 1). Note that, this process is opposed to the static per-image attention used in recent TTT-based architectures [23, 66].

3.2. Disentangled Memory Supervision

A naïve joint training of the cross-view attention module and the memory module is fundamentally underconstrained. Because disocclusion signals in streaming video are naturally sparse, the model tends to over-rely on the attention and effectively bypasses historical memory. To force the persistent internalization of scene context, we introduce an alternating training regime across a sequence of T timesteps.

Memory Supervision. As illustrated in Fig. 2, on even timesteps ($t = 0, 2, \dots$), we read out from the memory and supervise this readout to reconstruct a random target view. To enforce this loss on the memory alone, it is critical to ensure we read out purely from the memory, completely isolated from the current inputs. Therefore, we introduce additional *memory readout tokens*, representing the camera rays of the target view. Crucially, we restrict the attention mask of the aforementioned cross-view attention mod-

ule such that these memory tokens only self-attend amongst themselves. These isolated tokens then query the updated memory and pass through the feed-forward layers. Ultimately, they form an image, denoted as $\mathbf{I}_t^{\text{mem}}$, relying entirely on the internalized historical state. We supervise this blind retrieval via an auxiliary *memory loss*:

$$\mathcal{L}_{\text{mem}}^{(t)} = \mathcal{L}(\mathbf{I}_t^{\text{mem}}, \mathbf{I}_t^{\text{gt}}), \quad (5)$$

where $\mathcal{L}(\mathbf{I}, \mathbf{I}^{\text{gt}}) = \text{MSE}(\mathbf{I}, \mathbf{I}^{\text{gt}}) + \lambda_{\text{lpips}} \text{LPIPS}(\mathbf{I}, \mathbf{I}^{\text{gt}})$.

$\lambda_{\text{lpips}} = 0.5$ is the weight to balance the LPIPS loss [65].

Synthesis Supervision. On odd timesteps ($t = 1, 3, \dots$), the network operates in standard novel view synthesis mode and we supervise the synthesized target view. Here, the standard target ray tokens freely cross-attend with the tokenized inputs $\mathcal{O}_t$ while querying the persistent memory $\mathcal{W}_{t-1}$ inherited from the immediately preceding memorization step. Consequently, despite dynamic deformations between the current observation and the previous memory state, if a region is occluded, the network learns to dynamically fuse and recover it from this preceding memory. We supervise the output target image from this joint synthesis, denoted as $\mathbf{I}_t^{\text{synth}}$, using the identical loss formulation:

$$\mathcal{L}_{\text{synth}}^{(t)} = \mathcal{L}(\mathbf{I}_t^{\text{synth}}, \mathbf{I}_t^{\text{gt}}). \quad (6)$$

This alternating process continues until all T observations in the sequence are processed. The final training objective minimizes the joint loss across all timesteps, balancing the primary novel view synthesis task against the auxiliary memory regularization:

$$\mathcal{L}_{\text{total}} = \sum_{\substack{t=1 \\ t \text{ is odd}}}^T \mathcal{L}_{\text{synth}}^{(t)} + \sum_{\substack{t=0 \\ t \text{ is even}}}^T \mathcal{L}_{\text{mem}}^{(t)}, \quad (7)$$

where λ_{mem} controls the strength of the memory supervision.

Training-Inference Asymmetry. While we train with a 1:1 alternating ratio, our framework can execute significantly more synthesis steps than memory updates during inference. This discrepancy does not degrade performance because the synthesis steps between two memorization steps read out from the same frozen memory state and are completely independent of one another. See our supplemental videos where we synthesize 29 frames for every 1 step of memorization (29:1).

3.3. Memory Caching

TTT suffers from RNN drift over long-horizon updates. We observe this as memory decay, stemming from optimization instability, in the LaCT-NVS [66] column of Fig. 4. To address this problem, we adapt a memory caching strategy recently proposed for large language models [6, 7].

Method	4 Timesteps						30 Timesteps						60 Timesteps					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	29.13	0.9666	0.0258	0.1040	19.55	0.5790	28.98	0.9656	0.0263	0.1055	19.40	0.5707	28.99	0.9662	0.0257	0.1034	19.34	0.5670
LaCT-NVS	30.69	0.9741	0.0196	0.0822	21.72	0.7110	30.38	0.9708	0.0216	0.0885	21.22	0.6507	25.85	0.9500	0.0360	0.1236	16.25	0.5158
Token-Mem	29.36	0.9674	0.0249	0.1009	20.40	0.6409	28.36	0.9621	0.0290	0.1136	19.14	0.5624	28.11	0.9617	0.0295	0.1141	18.84	0.5505
Ours	30.43	0.9731	0.0203	0.0851	21.46	0.6971	30.03	0.9700	0.0225	0.0915	20.86	0.6434	29.99	0.9701	0.0225	0.0919	20.71	0.6286

Table 1. **Memory Stress Test Quantitative Comparisons.** The **best**, **second-best**, and **third-best** results are highlighted. We evaluate memory retention quality across 390 scenes (see Sec. 4.1), measured at timestamps $T = 4, 30, 60$. While our method maintains an mPSNR above 20 dB across all tested durations, LaCT-NVS drops by 5.47 dB and Token-Mem by 1.56 dB from $T = 4$ to $T = 60$. As a stateless model, LVSM fails to reconstruct details in the back view when only observing frontal views, and maintains a nearly flat performance across time. Please refer to our supplementary website for comprehensive video comparisons.

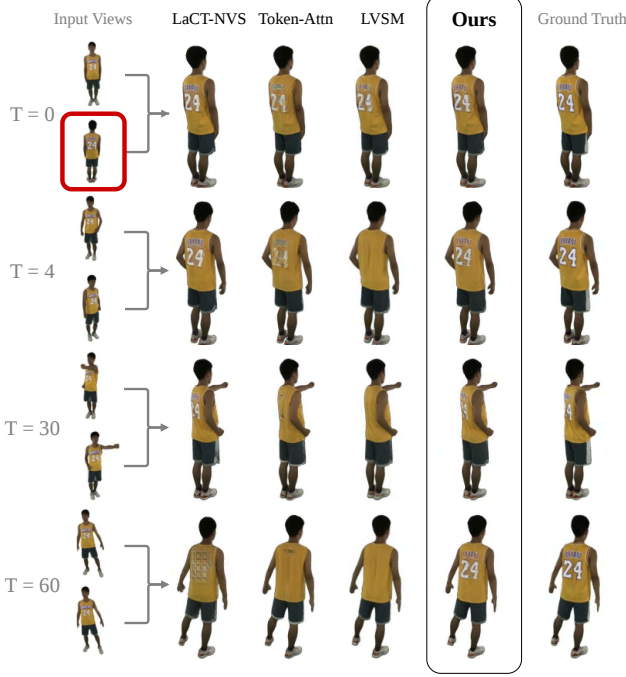


Figure 4. **Memory Stress Test Over Time.** We show the back view at $T=0$. For subsequent timestamps, we show only frontal views and task the model with synthesizing the occluded back. NSTM maintains high-fidelity recall over time while baseline models suffer from drift or collapse to stateless prior over time.

Concretely, the sequential application of our periodic memory update generates a sequence of evolving memory states $\mathcal{W}_0, \mathcal{W}_1, \mathcal{W}_2, \dots$, where the ones between updates are the same. Note, at any timestep t , the active memory is $\mathcal{W}_{t-1}$ since the memory is not reset after a snapshot is taken. We average over the active memory and every K past memory states, which can be implemented recursively.

$$\bar{\mathcal{W}}_0 = \mathcal{W}_0, \bar{\mathcal{W}}_n = \frac{n-1}{n}\bar{\mathcal{W}}_{n-1} + \frac{1}{n}\mathcal{W}_{nK}, \forall n > 1,$$

$$\hat{\mathcal{W}}_{t-1} = \frac{1}{m+1} \left(\mathcal{W}_{t-1} + \sum_{i=1}^m \mathcal{W}_{iK} \right) = \frac{\mathcal{W}_{t-1} + m\bar{\mathcal{W}}_m}{m+1}$$

where $m = \lfloor (t-1)/K \rfloor$. $\bar{\mathcal{W}}_m$ is the average of every K past states. In practice, we only keep one extra set of weights $\bar{\mathcal{W}}_m$ to track the whole average. This aggregated state $\hat{\mathcal{W}}_{t-1}$ replaces standard memory conditioning (Eq. 1), stabilizing the parameter space and mitigating the out-of-

distribution drift caused by continuous gradient updates. Importantly, this memory caching strategy fails on standard, coupled TTT architectures. Its effectiveness fundamentally relies on our cross-view attention to explicitly handle spatial deformations, as we empirically validate in Sec. 4.3.

4. Experiments

We validate NSTM’s ability to maintain real-time, high-fidelity, minute-long memory under severe occlusions. We detail our experimental setup in Sec. 4.1, evaluate memory persistence and synthesis fidelity in Sec. 4.2, deconstruct our architecture via an ablation study in Sec. 4.3, and analyze our amortized efficiency in Sec. 4.4. We encourage readers to view our supplementary videos for convincing comparisons.

4.1. Experimental Setup

Dataset. We evaluate on MVHumanNet++ [28], featuring 60,000 multi-view motion sequences across 4,500 identities and 9,000 outfits, using a 90%/10% train/test split. The dataset’s wide view coverage, unconstrained motion and minute-long sequence length make it an ideal testbed for dynamic NVS and long-term memory tasks.

Evaluation Protocol. MVHumanNet++ [28] provides annotated train/eval sets of 60-frame sequences (subsampled at 1.2 FPS from 50s of raw video), enabling minute-scale evaluation. However, measuring long-term memorization in the wild is challenging as natural disocclusions are spatially and temporally sparse – only 17 of 390 test scenes exhibit full rotations that can best test memorization. To comprehensively evaluate memory, we use two protocols. (1) *Memory Stress Test*: A forced-occlusion setup to evaluate memorization across all scenes. The model observes full front and back views at $T = 0$. For subsequent frames, we restrict inputs to frontal stereo views and task the model with synthesizing the occluded back. Evaluations at 4, 30, and 60 frames quantify memory decay from short-term horizons to the minute-scale limit. (2) *Memory from Natural Rotations*: For the 17 full-rotation scenes, we provide fixed frontal stereo inputs and evaluate synthesis as the back naturally reveals and occludes. To avoid the backgrounds dominating the metrics, we report masked mPSNR and mSSIM isolated over the dynamic foreground. All evaluation are done at 256×256 resolution on one H100 GPU. Results

for the Memory Stress Test are presented in Sec. 4.2 and results for the Memory from Natural Rotations and an unconstrained NVS setup are shown in Sec. A.

Baselines. We compare against the stateless LVSM [22] and two stateful baselines: fast-weight-based LaCT-NVS [66] (adapted to causal updates for dynamic scenes) and token-memory-based Token-Mem (an NVS adaptation of CUT3R [49]). See Sec. B.3 for further adaption details on baseline models. Following their original designs, both stateful methods update memory per frame. Based on our memorization-synthesis decoupling, NSTM updates memory every two frames. Note our subsampled video test cases actually artificially benefit the stateful baselines by mitigating their rapid drift at higher frame rates. In contrast, NSTM’s results accurately reflect true performance on normal high-FPS videos as its dense synthesis steps operate independently between these periodic memorization updates.

Implementation details. NSTM and all baselines are implemented in JAX and trained with identical optimizer hyperparameters and a batch size of 128. All 256×256 stateful models adopt a two-stage training curriculum: memory bootstrapping pretraining on 4-frame clips (64 H100 GPUs), followed by fine-tuning on 24-frame clips (128 H100 GPUs) for free-viewpoint synthesis and long-term memory. The 512×512 model undergoes an additional 24-frame fine-tuning at 512×512 using 128 B200 GPUs. The stateless LVSM[22] adopts only one stage using one frame with 32 H100 GPUs. See Sec. B.1 for full details. Inference latency is evaluated on a single H100 GPU at 256×256 . We set $\lambda_{\text{LPIPS}} = 0.5$.

4.2. Memory Stress Test

We present our quantitative comparisons in Tab. 1 and qualitative visual evidence in Fig. 3. In the short term ($T = 4$), the advantage of explicit memory is clear: NSTM, LaCT-NVS, and Token-Mem all outperform the stateless LVSM, which struggles without historical context. However, the fragility of prior stateful methods becomes visible over the minute-long context ($T = 60$). Token-Mem quickly forgets historical context, its performance comparable to the stateless LVSM that never observed the back view. LaCT-NVS suffers a 5.47 dB drop in PSNR due to the instability of its continuous update mechanism. This quantitative collapse is clearly visible in Fig. 3 and Fig. 4: where LaCT-NVS suffers from out-of-distribution (OOD) drift, while LVSM produces blurry outputs in unseen regions (e.g. the back) due to its stateless nature.

On the other hand, NSTM maintains highly robust long-term memory which is especially pronounced in our masked foreground metrics (mPSNR/mSSIM), proving the model accurately locks in the dynamic subject’s identity rather than relying on the background. Qualitatively (Fig. 3), NSTM faithfully reconstructs specific, temporarily

Method	Ablation Study (over 60 timesteps)				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
LaCT-NVS Baseline	22.32	0.9275	0.0616	13.49	0.4942
LaCT-NVS w/ L_2	27.77	0.9594	0.0275	19.08	0.6189
LaCT-NVS w/ L_2 w/ Mem Caching	20.25	0.9135	0.0688	12.03	0.4707
Ours w/ $\mathcal{L}_{\text{mem}}$	28.23	0.9639	0.0251	19.36	0.6198
Ours w/ Mem Caching	29.24	0.9673	0.0232	19.78	0.6321
Ours w/ Mem Caching	30.09	0.9705	0.0217	20.79	0.6537

Table 2. **Quantitative Ablation Study.** best, second best, and third best are colored. Metrics are averaged over 60 timesteps over 390 MVHumanNet++ [55] eval scenes.

occluded details—such as complex logo and hair styles—long after they disappear from the input stream. This sustained high-fidelity recall directly validates the stabilizing effect of our L_2 objective and memory caching strategy.

4.3. Ablation Study

We validate our design choices in Tab. 2 and Fig. 5, measuring the impact of each component on the pretrained models using the 60-frame memory stress test protocol.

Stability via L_2 Objective. We first establish a naïve causal baseline (LaCT-NVS) [66] that updates memory at every frame without decoupling memorization from synthesis. As observed, the unbounded dot-product inner loss causes magnitude inflation over continuous updates, resulting in poor performance (22.32 PSNR). Replacing this with a stable L_2 objective (LaCT-NVS w/ L_2) significantly improves stability and synthesis quality (+5.45 PSNR).

Memory Caching Requires Decoupled Training. While memory caching is designed for long-term stability, it fundamentally fails on coupled architectures. Row 3 evaluates the LaCT-NVS with L_2 baseline by performing its standard update-and-apply step at every frame, but uniquely conditioning it on the aggregated memory cache (Sec. 3.3). This causes a severe performance drop (20.25 PSNR). The synthesized output becomes visually “stuck” in a historical pose (see LaCT-NVS w/ L_2 w/ caching column in Fig. 5). This occurs because temporal motion is inherently baked into the fast-weights; aggregating these weights creates a superposition of past poses. In our decoupled architecture (Sec. 3.1), the cross-view attention explicitly learns to resolve and deform from this superimposed historical memory to match the current observation. The tightly coupled baseline lacks this explicit deformation mechanism, causing it to simply output the averaged past poses. This proves that structurally decoupling memorization from synthesis is necessary to successfully aggregate historical memory.

Disentangled Supervision Enhances Memory Persistence. When we fully decouple the architecture but remove the auxiliary memory loss $\mathcal{L}_{\text{mem}}$ (Ours w/ $\mathcal{L}_{\text{mem}}$), performance drops to 28.23 PSNR. Without explicit pressure to reconstruct the scene independently, the network takes a shortcut, over-relying on the cross-view attention’s current

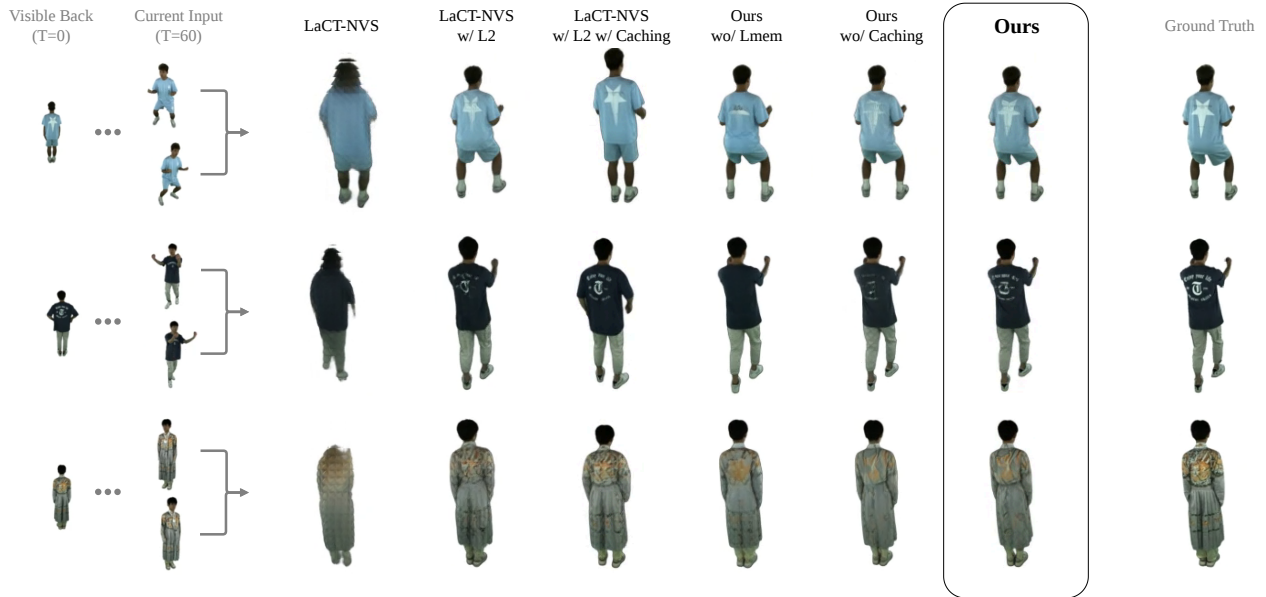


Figure 5. **Qualitative ablation study.** At $T = 60$, each model synthesizes the back view that has been occluded after $T = 0$. **LaCT-NVS** demonstrates the instability of the inner dot-product loss. In contrast, **LaCT-NVS w/ L_2** greatly improves stability through our L_2 inner loss choice, though it still suffers from some drift. **LaCT-NVS w/ L_2 w/ Caching** freezes into an averaged past pose, demonstrating that our memorization-synthesis decoupling with cross-view attention is essential for effective memory caching. Dropping $\mathcal{L}_{\text{mem}}$ from our method collapses recall, validating our disentangled memory supervision design (Sec. 3.2). Furthermore, dropping memory caching reduces the recall horizon. Only the full model (**Ours**) provides minute-scale recall of the historical back view while capturing current motion. See our supplementary website for the video results.

visual context and partially collapsing to a stateless prior. Introducing $\mathcal{L}_{\text{mem}}$ forces robust geometric internalization (+1.01 PSNR).

Long-term Stability via Memory Caching. Finally, applying our memory caching strategy to the fully decoupled and properly supervised model (Ours w/ Mem Caching) yields our highest performance (30.09 PSNR), successfully mitigating long-term weight drift and preserving minute-long context.

4.4. Real-Time Inference Efficiency

To validate our claim of real-time, long-horizon generation, we compare the amortized inference latency of NSTM against *Temporal-LVSM*, an adaptation of LVSM [22] for dynamic video sequences. To handle long contexts in this baseline, we compute dense attention across all past observations that we want to memorize. As shown in Fig. 1, this imposes a quadratic computational cost, causing latency to spike and precluding real-time rendering on continuous streams. In contrast, NSTM architecturally bypasses this bottleneck, achieving constant inference latency regardless of video length. Our decoupling strategy breaks the tie between temporal horizon and computational cost, unlocking sustained, real-time dynamic view synthesis.

5. Conclusion and Limitations

We presented Neural Space-Time Memory (NSTM), a novel online dynamic NVS framework operating on multi-

view streaming videos. By disentangling periodic memory updates from per-frame transformer-based deformation modeling and introducing a targeted memory auxiliary loss, NSTM internalizes persistent scene context while maintaining amortized real-time speed. Furthermore, our memory caching strategy and stable L_2 update rule unlock minute-long memory retention, achieving state-of-the-art results.

While NSTM provides a principled solution for streaming dynamic NVS, we identify several avenues for future research. *Update Frequency and Missed Events:* Periodic updates may miss events between memorization steps. Large-chunk updates that memorize multiple frames at once, each with its own time embedding, could mitigate this and additionally support irregular frame spacing (e.g., unsynced multi-view rigs), leaving open the tradeoff between update frequency and batch size. *Finite Memory Capacity:* Models like ours eventually hit capacity limits; future iterations could manage memory caching weights in a learned way. *Memory Primacy Bias:* The method favors memorization during early steps; larger-scale training on longer sequences and better mid-sequence frame sampling could encourage long-term retention. *Complex Motion and Trajectories:* Resolving long-range camera and subject motion within the compressed memory remains challenging and is left for future work.

6. Acknowledgements

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References

- [1] Kara-Ali Aliev, Artem Sevastopolsky, Maria Kolos, Dmitry Ulyanov, and Victor Lempitsky. Neural point-based graphics. In *European conference on computer vision*, pages 696–712. Springer, 2020. 2
- [2] Simran Arora, Sabri Eyuboglu, Michael Zhang, Aman Timalisina, Silas Alberti, James Zou, Atri Rudra, and Christopher Re. Simple linear attention language models balance the recall-throughput tradeoff. In *International Conference on Machine Learning*, pages 1763–1840. PMLR, 2024. 3
- [3] Ali Behrouz, Zeman Li, Praneeth Kacham, Majid Daliri, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, and Vahab Mirrokni. Atlas: Learning to optimally memorize the context at test time, 2025. 2, 3, 4
- [4] Ali Behrouz, Meisam Razaviyayn, Peilin Zhong, and Vahab Mirrokni. Nested learning: The illusion of deep learning architectures. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 3
- [5] Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. 3
- [6] Ali Behrouz, Zeman Li, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, and Vahab Mirrokni. Memory caching: Rnns with growing memory, 2026. 2, 3, 5
- [7] Ali Behrouz, Meisam Razaviyayn, Peilin Zhong, and Vahab Mirrokni. It’s all connected: A journey through test-time memorization, attentional bias, retention, and online optimization. In *The Fourteenth International Conference on Learning Representations*, 2026. 2, 3, 4, 5
- [8] Michael Broxton, John Flynn, Ryan Overbeck, Daniel Erickson, Peter Hedman, Matthew Duvall, Jason Dourgarian, Jay Busch, Matt Whalen, and Paul Debevec. Immersive light field video with a layered mesh representation. *ACM Transactions on Graphics (TOG)*, 39(4):86–1, 2020. 2
- [9] Ang Cao and Justin Johnson. Hexplane: A fast representation for dynamic scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 130–141, 2023. 3
- [10] Keyan Ding, Kede Ma, Shiqi Wang, and Eero P. Simoncelli. Image quality assessment: Unifying structure and texture similarity. *CoRR*, abs/2004.07728, 2020. 12
- [11] Xin Dong, Yonggan Fu, Shizhe Diao, Wonmin Byeon, ZIJIA CHEN, Ameya Sunil Mahabaleshwarkar, Shih-Yang Liu, Min-Hung Chen, Yoshi Suhara, Yingyan Celine Lin, et al. Hymba: A hybrid-head architecture for small language models. In *The Thirteenth International Conference on Learning Representations*. 3
- [12] John Flynn, Ivan Neulander, James Philbin, and Noah Snavely. Deepstereo: Learning to predict new views from the world’s imagery. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5515–5524, 2016. 2
- [13] John Flynn, Michael Broxton, Paul Debevec, Matthew DuVall, Graham Fyffe, Ryan Overbeck, Noah Snavely, and Richard Tucker. Deepview: View synthesis with learned gradient descent. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2367–2376, 2019.
- [14] John Flynn, Michael Broxton, Lukas Murmann, Lucy Chai, Matthew DuVall, Clément Godard, Kathryn Heal, Srinivas Kaza, Stephen Lombardi, Xuan Luo, et al. Quark: Real-time, high-resolution, and general neural view synthesis. *ACM Transactions on Graphics (TOG)*, 43(6):1–20, 2024. 2
- [15] Sara Fridovich-Keil, Giacomo Meanti, Frederik Rahbæk Warburg, Benjamin Recht, and Angjoo Kanazawa. K-planes: Explicit radiance fields in space, time, and appearance. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12479–12488, 2023. 3
- [16] Han Guo, Songlin Yang, Tarushii Goel, Eric P Xing, Tri Dao, and Yoon Kim. Log-linear attention. *arXiv preprint arXiv:2506.04761*, 2025. 3
- [17] Ramin Hasani, Mathias Lechner, Tsun-Hsuan Wang, Makram Chahine, Alexander Amini, and Daniela Rus. Liquid structural state-space models. In *The Eleventh International Conference on Learning Representations*. 3
- [18] Peter Hedman, Julien Philip, True Price, Jan-Michael Frahm, George Drettakis, and Gabriel Brostow. Deep blending for free-viewpoint image-based rendering. *ACM Transactions on Graphics (ToG)*, 37(6):1–15, 2018. 2
- [19] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020. 2
- [20] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2d gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH 2024 conference papers*, pages 1–11, 2024. 2
- [21] Kazuki Irie, Francesco Faccio, and Jürgen Schmidhuber. Neural differential equations for learning to program neural nets through continuous learning rules. *Advances in Neural Information Processing Systems*, 35:38614–38628, 2022. 3
- [22] Haian Jin, Hanwen Jiang, Hao Tan, Kai Zhang, Sai Bi, Tianyuan Zhang, Fujun Luan, Noah Snavely, and Zexiang Xu. Lvsm: A large view synthesis model with minimal 3d inductive bias. In *The Thirteenth International Conference on Learning Representations*, 2025. 1, 2, 4, 7, 8, 12, 13, 14, 15
- [23] Haian Jin, Rundui Wu, Tianyuan Zhang, Ruiqi Gao, Jonathan T. Barron, Noah Snavely, and Aleksander Holynski. ZipMap: Linear-time stateful 3d reconstruction via test-time training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2, 5

- [24] Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks, 2024. 4
- [25] Praneeth Kacham, Vahab Mirrokni, and Peilin Zhong. Polysketchformer: Fast transformers via sketching polynomial kernels. In *International Conference on Machine Learning*, pages 22748–22770. PMLR, 2024. 3
- [26] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning*, pages 5156–5165. PMLR, 2020. 3
- [27] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, George Drettakis, et al. 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.*, 42(4):139–1, 2023. 2
- [28] Chenghong Li, Hongjie Liao, Yihao Zhi, Xihe Yang, Zhengwentai Sun, Jiahao Chang, Shuguang Cui, and Xiaoguang Han. Mvhumannet++: A large-scale dataset of multi-view daily dressing human captures with richer annotations for 3d human digitization. *arXiv preprint arXiv:2505.01838*, 2025. 6
- [29] Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim, Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, et al. Neural 3d video synthesis from multi-view video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5521–5531, 2022. 3
- [30] Zhan Li, Zhang Chen, Zhong Li, and Yi Xu. Spacetime gaussian feature splatting for real-time dynamic view synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8508–8520, 2024. 3
- [31] Haotong Lin, Sida Peng, Zhen Xu, Yunzhi Yan, Qing Shuai, Hujun Bao, and Xiaowei Zhou. Efficient neural radiance fields for interactive free-viewpoint video. In *SIGGRAPH Asia 2022 conference papers*, pages 1–9, 2022. 2
- [32] Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot one image to 3d object. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9298–9309, 2023. 2
- [33] Yuan Liu, Cheng Lin, Zijiao Zeng, Xiaoxiao Long, Lingjie Liu, Taku Komura, and Wenping Wang. Syncdreamer: Generating multiview-consistent images from a single-view image. *arXiv preprint arXiv:2309.03453*, 2023. 2
- [34] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021. 2
- [35] Bo Peng, Ruichong Zhang, Daniel Goldstein, Eric Alcaide, Xingjian Du, Haowen Hou, Jiaju Lin, Jiaxing Liu, Janna Lu, William Merrill, Guangyu Song, Kaifeng Tan, Saiteja Utpala, Nathan Wilce, Johan S. Wind, Tianyi Wu, Daniel Wutke, and Christian Zhou-Zheng. RWKV-7 “goose” with expressive dynamic state evolution. In *Second Conference on Language Modeling*, 2025. 3
- [36] Gernot Riegler and Vladlen Koltun. Free view synthesis. In *European conference on computer vision*, pages 623–640. Springer, 2020. 2
- [37] Mehdi SM Sajjadi, Henning Meyer, Etienne Pot, Urs Bergmann, Klaus Greff, Noha Radwan, Suhani Vora, Mario Lučić, Daniel Duckworth, Alexey Dosovitskiy, et al. Scene representation transformer: Geometry-free novel view synthesis through set-latent scene representations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6229–6238, 2022. 2
- [38] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International conference on machine learning*, pages 9355–9366. PMLR, 2021. 3
- [39] Yichun Shi, Peng Wang, Jianglong Ye, Mai Long, Kejie Li, and Xiao Yang. Mvdream: Multi-view diffusion for 3d generation. *arXiv preprint arXiv:2308.16512*, 2023. 2
- [40] Vincent Sitzmann, Michael Zollhöfer, and Gordon Wetzstein. Scene representation networks: Continuous 3d-structure-aware neural scene representations. *Advances in neural information processing systems*, 32, 2019. 2
- [41] Jiakai Sun, Han Jiao, Guangyuan Li, Zhanjie Zhang, Lei Zhao, and Wei Xing. 3dstream: On-the-fly training of 3d gaussians for efficient streaming of photo-realistic free-viewpoint videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20675–20685, 2024. 3
- [42] Yutao Sun, Li Dong, Shaohan Huang, Shuming Ma, Yuqing Xia, Jilong Xue, Jiaoyong Wang, and Furu Wei. Retentive network: A successor to transformer for large language models. *arXiv preprint arXiv:2307.08621*, 2023. 3
- [43] Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): Rnns with expressive hidden states, 2025. 3, 4
- [44] Justus Thies, Michael Zollhöfer, and Matthias Nießner. Deferred neural rendering: Image synthesis using neural textures. *Acm Transactions on Graphics (TOG)*, 38(4):1–12, 2019. 2
- [45] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. 3
- [46] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 5294–5306, 2025. 2
- [47] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. Neus: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. *arXiv preprint arXiv:2106.10689*, 2021. 2
- [48] Qianqian Wang, Zhicheng Wang, Kyle Genova, Pratul P Srinivasan, Howard Zhou, Jonathan T Barron, Ricardo Martin-Brualla, Noah Snavely, and Thomas Funkhouser. Ibrnet: Learning multi-view image-based rendering. In *Pro-*

- ceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 4690–4699, 2021. 2
- [49] Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A. Efros, and Angjoo Kanazawa. Continuous 3D Perception Model with Persistent State. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10510–10522, Los Alamitos, CA, USA, 2025. IEEE Computer Society. 7, 12, 14
- [50] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20697–20709, 2024. 2
- [51] Yiming Wang, Qin Han, Marc Habermann, Kostas Daniilidis, Christian Theobalt, and Lingjie Liu. Neus2: Fast learning of neural implicit surfaces for multi-view reconstruction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3295–3306, 2023. 2
- [52] Yiming Wang, Lucy Chai, Xuan Luo, Michael Niemeyer, Manuel Lagunas, Stephen Lombardi, Siyu Tang, and Tiancheng Sun. Learning efficient fuse-and-refine for feed-forward 3d gaussian splatting. *arXiv preprint arXiv:2503.14698*, 2025. 3
- [53] Daniel Watson, William Chan, Ricardo Martin-Brualla, Jonathan Ho, Andrea Tagliasacchi, and Mohammad Norouzi. Novel view synthesis with diffusion models. *arXiv preprint arXiv:2210.04628*, 2022. 2
- [54] Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4d gaussian splatting for real-time dynamic scene rendering. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20310–20320, 2024. 2, 3
- [55] Zhangyang Xiong, Chenghong Li, Kenkun Liu, Hongjie Liao, Jianqiao Hu, Junyi Zhu, Shuliang Ning, Lingteng Qiu, Chongjie Wang, Shijie Wang, Shuguang Cui, and Xiaoguang Han. Mvhumannet: A large-scale dataset of multi-view daily dressing human captures. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19801–19811, 2024. 2, 7
- [56] Qiangeng Xu, Zexiang Xu, Julien Philip, Sai Bi, Zhixin Shu, Kalyan Sunkavalli, and Ulrich Neumann. Point-nerf: Point-based neural radiance fields. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5438–5448, 2022. 2
- [57] Zhen Xu, Sida Peng, Haotong Lin, Guangzhao He, Jiaming Sun, Yujun Shen, Hujun Bao, and Xiaowei Zhou. 4k4d: Real-time 4d view synthesis at 4k resolution. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20029–20040, 2024. 3
- [58] Zhen Xu, Zhengqin Li, Zhao Dong, Xiaowei Zhou, Richard Newcombe, and Zhaoyang Lv. 4dgt: Learning a 4d gaussian transformer using real-world monocular videos. *arXiv preprint arXiv:2506.08015*, 2025. 2
- [59] Jinbo Yan, Rui Peng, Zhiyan Wang, Luyang Tang, Jiayu Yang, Jie Liang, Jiahao Wu, and Ronggang Wang. Instant gaussian stream: Fast and generalizable streaming of dynamic scene reconstruction via gaussian splatting. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 16520–16531, 2025. 3
- [60] Ziyi Yang, Xinyu Gao, Wen Zhou, Shaohui Jiao, Yuqing Zhang, and Xiaogang Jin. Deformable 3d gaussians for high-fidelity monocular dynamic scene reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 20331–20341, 2024. 3
- [61] Lior Yariv, Yoni Kasten, Dror Moran, Meirav Galun, Matan Atzmon, Basri Ronen, and Yaron Lipman. Multiview neural surface reconstruction by disentangling geometry and appearance. *Advances in Neural Information Processing Systems*, 33:2492–2502, 2020. 2
- [62] Lior Yariv, Jiatao Gu, Yoni Kasten, and Yaron Lipman. Volume rendering of neural implicit surfaces. *Advances in neural information processing systems*, 34:4805–4815, 2021. 2
- [63] Wangbo Yu, Jinbo Xing, Li Yuan, Wenbo Hu, Xiaoyu Li, Zhipeng Huang, Xiangjun Gao, Tien-Tsin Wong, Ying Shan, and Yonghong Tian. Viewcrafter: Taming video diffusion models for high-fidelity novel view synthesis. *arXiv preprint arXiv:2409.02048*, 2024. 2
- [64] Junyi Zhang, Charles Herrmann, Junhwa Hur, Chen Sun, Ming-Hsuan Yang, Forrester Cole, Trevor Darrell, and Deqing Sun. Loger: Long-context geometric reconstruction with hybrid memory. *arXiv preprint arXiv:2603.03269*, 2026. 2
- [65] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. *CoRR*, abs/1801.03924, 2018. 5
- [66] Tianyuan Zhang, Sai Bi, Yicong Hong, Kai Zhang, Fujun Luan, Songlin Yang, Kalyan Sunkavalli, William T Freeman, and Hao Tan. Test-time training done right. *arXiv preprint arXiv:2505.23884*, 2025. 2, 3, 4, 5, 7, 12, 14

A. Additional Experiments

A.1. Memory from Natural Rotations Test

To evaluate NSTM on the general setting of natural video sequences where occlusions can happen naturally at any point, we evaluate 17 scenes in MVHumanNet++ that feature back reveal during natural rotations. This is in contrast to our “Memory Stress Test” Tab. 1 where the model sees the back view only at timestep 0 and needs to remember it while receiving only frontal views over the entire duration of the video. To test, we provide fixed frontal stereo inputs (cam15 and cam01) and evaluate synthesis as the back naturally reveals and occludes.

Results and Discussions. We present our quantitative comparisons in Tab. 5 and qualitative visual evidence in Fig. 7 measuring performance both averaged across all frames and on the final frame in the ($t = 60$) sequence. The strength of NSTM remains evident, outperforming Token-Mem and LaCT-NVS by 1.74 dB and 2.95 dB mP-SNR on last frame measurement. Since LVSM is a stateless

Method	Novel View Synthesis Over 60 Timesteps					
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	DISTS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
LVSM	29.48	0.9699	0.0225	0.0917	20.10	0.6509
LaCT-NVS	26.84	0.9567	0.0281	0.0997	17.62	0.5869
Token-Mem	22.81	0.9273	0.0477	0.1180	14.04	0.4595
Ours	29.11	0.9696	0.0223	0.0894	19.93	0.6703

Table 3. **General Novel View Synthesis Comparison.** **best**, **second best**, and **third best** are highlighted. Given stereo inputs from fixed frontal cameras, all models synthesize arbitrary unseen viewpoints over time. The metrics are averaged over 60 timesteps across 390 evaluation scenes. NSTM significantly outperforms prior stateful baselines (LaCT-NVS [66] and Token-Mem [49]) by successfully mitigating long-term memory degradation. Because the sparsity of disocclusions in this protocol limits the benefits of historical context, NSTM performs on par with the stateless LVSM baseline [22], confirming its strong general NVS capabilities. See videos in our supplementary website for qualitative visual results.

model the average and last frame performance are almost identical, demonstrating that it is not able to remember past details that would allow it to increase its accuracy.

A.2. General Novel View Synthesis Test

In addition to the memory stress test in Sec. 4.2 and natural memory test in Sec. A.1, NSTM is fundamentally designed to be a highly versatile dynamic NVS framework. To evaluate its generalized synthesis capabilities, we conduct experiments under a standard, unconstrained streaming protocol. **Evaluation Protocol.** To simulate typical real-world telepresence or broadcasting scenarios where input camera rigs are fixed, we restrict the streaming inputs to only two static frontal views (`cam15` and `cam01`). The model is then tasked with rendering the dynamic scene from arbitrary time-varying target viewpoints. Because disocclusions under this protocol are spatially and temporally sparse, the benefit of long-term memory is naturally limited. Thus, we use this setup strictly to isolate and stress-test the model’s spatial generalization – specifically, its ability to synthesize arbitrary unseen viewpoints from instantaneous inputs.

Results and Discussions. Despite temporal context provides limited benefits, Table 3 shows that NSTM consistently outperforms the stateful LaCT-NVS and Token-Mem baselines. Furthermore, NSTM remains competitive in standard structural metrics (LPIPS, SSIM, and mSSIM) against the purely stateless LVSM baseline, while outperforming on LPIPS and DISTS [10]. This demonstrates our model’s spatial rendering capabilities.

A.3. Inference Speed

As mentioned in the Introduction, the memory update and apply operations exhibit asymmetric computational costs. As detailed in Tab. 4, a full memorization step (both update and apply) requires 58.14 ms, whereas a synthesis step (apply only) takes just 27.01 ms. Consequently, by scheduling memorization at 1 FPS and synthesis at 30 FPS, our frame-

Step	Latency (ms) $\downarrow$	FPS $\uparrow$
Memorization	58.14	17.19
Synthesis	27.01	37.02

Table 4. **Computation Asymmetry.** The memorization step (including both memory update and apply) is significantly slower than the synthesis step (apply only). Inference performance breakdown measured on a single H100 GPU with a resolution of 256x256.

work achieves an amortized rendering speed of 28.1 ms per frame (see Fig. 1 of the main paper). All measurements were conducted at 256×256 resolution on a single NVIDIA H100 GPU.

B. Additional Implementation Details

In this section, we provide comprehensive details regarding our training curriculum, optimization hyperparameters, baseline configurations, and the evaluation protocol for our generalized Novel View Synthesis experiments.

B.1. Training Curriculum

All 256×256 stateful models are trained with a two-stage training curriculum. Pretraining bootstraps our model’s memory and is carried out on short video clips consisting of 4 frames. We then finetune for *free-viewpoint synthesis* and *long memory persistence* with long 24-frame video clips. To demonstrate our memory retention capabilities at 512×512 , we further finetune our 256×256 model in stage 3 on 512×512 videos. At each timestep, the network processes 2 sparse input views to synthesize 1 target view. Details on each stage follows:

Stage 1: Memory Bootstrap Pretraining. The objective of this initial phase is to explicitly encourage the network to build and rely upon its memory module. For the first frame of the sequence ($t = 0$), the input consists of one frontal view (`cam00`) and one back-facing view (`cam08`), providing full geometric context. For subsequent frames ($t > 0$), the inputs transition to two random views. Crucially, the target supervision is a target view that is explicitly distinct from the $t > 0$ inputs. Note these target viewpoints remain the same across the sequence. This rigid setup encourages memorizing the initial context to faithfully inpaint the targets at later timesteps.

Stage 2: Free-Viewpoint and Long Memory Adaptation Finetuning. Once the memory module is bootstrapped, we adapt the network to encourage longer memory persistence and enable synthesis of arbitrary, time-varying viewpoints. We extend the pretraining sequence from 4 frames to 24 frames and supervise the network at each timestep by a random, time-varying target view strictly withheld from the input stream. To construct the inputs, two random base

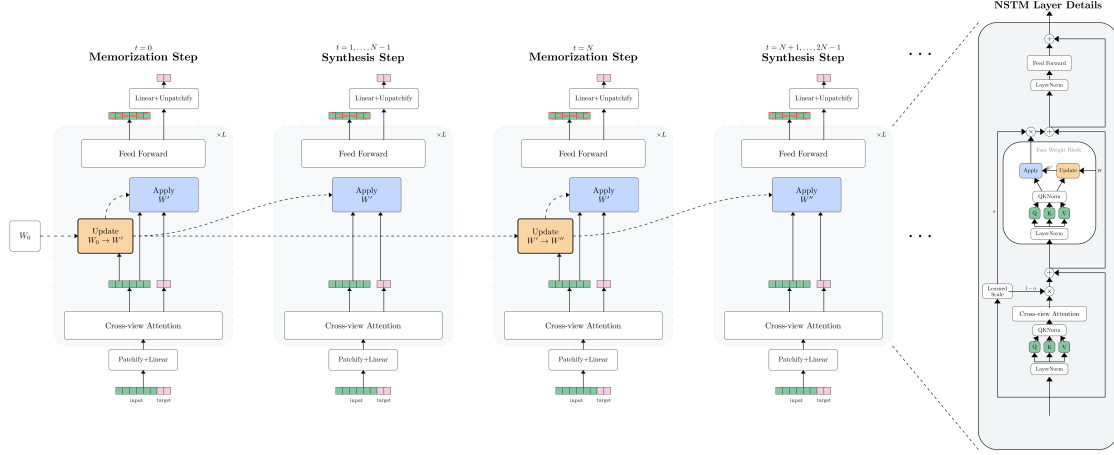


Figure 6. **Online Inference Pipeline and NSTM Layer Design.** **Left:** During inference, we run the memorization step periodically while performing the synthesis step per-frame, which enables us to achieve amortized real-time speed. **Right:** We introduce the NSTM Layer consisting of cross-view attention, a fast-weight block and a feed-forward layer. We also incorporate a learned scale that allows the network to balance the contribution of memory and attention to each layer’s output.

Method	Average over T=60						Last frame					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	28.81	0.9654	0.0286	0.1216	19.21	0.5069	28.81	0.9656	0.0296	0.1298	19.69	0.5272
LaCT-NVS	29.13	0.9667	0.0253	0.1045	19.98	0.5850	27.01	0.9551	0.0340	0.1414	17.93	0.4982
Token-Mem	27.24	0.9565	0.0353	0.1303	18.05	0.4926	28.11	0.9615	0.0329	0.1363	19.14	0.5194
Ours	29.53	0.9692	0.0247	0.1056	20.39	0.5928	29.68	0.9691	0.0250	0.1107	20.88	0.5997

Table 5. **Memory from Natural Rotations Test.** Best, second best, and third best are highlighted. Memory is critical for this evaluation protocol: the sequences feature natural rotations that reveal the back of the subject, which the models must accurately synthesize. Because the memory of stateful baselines degrades over time, NSTM significantly outperforms Token-Mem and LaCT-NVS by 1.74 dB and 2.95 dB in mPSNR, respectively, when measured on the final frame. Furthermore, by also surpassing the stateless LVSM [22], NSTM clearly demonstrates the advantage of our robust memory persistence. See our supplementary website for video results.

cameras are selected and held constant for all $t > 0$. At the initial timestep ($t = 0$), we enforce a forced-occlusion scenario by exclusively sourcing inputs from two distinct, alternative cameras. Finally, to ensure the model properly learns object-centric rotation rather than just camera translation, 50% of the training sequences undergo coordinate alignment. Specifically, we modify the extrinsics of the first initial $t = 0$ camera to match the first base view. Throughout this extended sequence, we maintain the alternating memorization and synthesis steps utilized in the pre-training phase.

Stage 3: High-Resolution (512×512) Finetuning. We further finetune at 512×512 resolution for 20K iterations. Similar to Stage 2, we simulate memory assimilation by injecting two alternative cameras. However, we inject the two alternative cameras at a random memorization step (even steps) rather than strictly at $t = 0$. To ensure sufficient subsequent frames remain for supervision, t is sampled from a

decaying distribution:

$$p(t) = \begin{cases} \frac{\gamma^{t/2}}{\sum_{j=0}^{T/2-1} \gamma^j} & \text{if } t \bmod 2 = 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\gamma = 0.5$, and $T = 24$ denotes the video sequence length.

B.2. Training Hyperparameters

We implement our framework using JAX. Models are trained with a total batch size of 128 for 500,000 iterations during the pretraining stage, followed by an additional 25,000 iterations during the finetuning stage. The 512×512 model undergoes another stage of high-resolution finetuning as in Sec. B.1. We employ a cosine learning rate decay schedule with a linear warmup. During pretraining, the learning rate is initialized at 1×10^{-6} , warms up over 4,000 steps to a peak of 4×10^{-4} , remains constant for 100,000 steps, and then decays down to a final value of 1×10^{-6} .

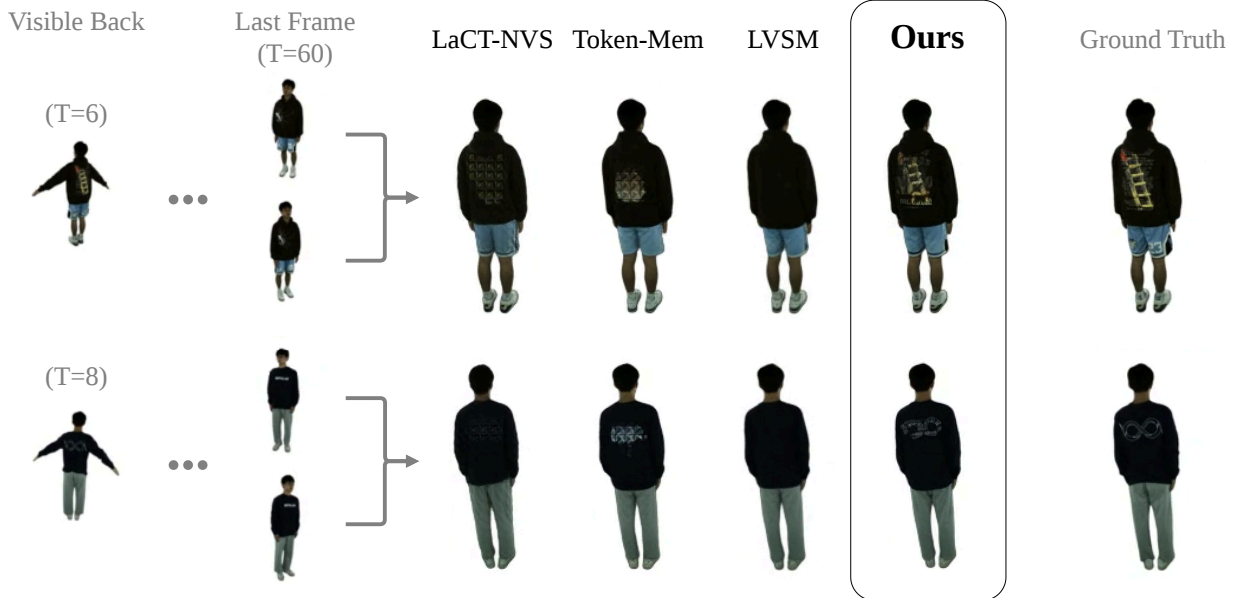


Figure 7. **Memory from Natural Rotations Qualitative Results.** We evaluate our method against the baselines on sequences feature natural rotations that reveal the back of the subject, which the models must synthesize. Left column shows at which timestep the back is revealed. NSTM recalls the clothing graphics. The stateless LVSM cannot synthesize the unseen graphics, while LaCT-NVS and Token-Mem show a degraded memory. See supplementary website for more video results.

During the finetuning stage, we retain the same schedule parameters but reduce the peak learning rate to 4×10^{-5} warming up over 4,000 steps and then remaining constant. Finally, for the image reconstruction objective across all models and stages, the perceptual loss weight is fixed at $\lambda_{\text{lpips}} = 0.5$.

B.3. Baseline Adaptations and Coordinate Alignments

To ensure a fair comparison, all baselines evaluated in Tab. 1 are of 256×256 resolution and all stateful models are trained on the exact same two-training curriculum. The LVSM baseline [22] is stateless and thus goes through only one pretraining stage as below. Due to architectural differences, camera coordinate systems must be aligned differently:

LVSM [22]: We utilize the exact architecture described in the original LVSM paper. But for fair comparison, we use the same learning rate schedule as above and the same L2 and LPIPS loss. Because LVSM is a stateless, per-frame model, we construct its training batches by extracting only the final frame ($t = 3$) from our 4-timestep pretraining video clips, ensuring it trains on identical data distributions. For coordinate alignment, we relativize the camera coordinate system to each individual target camera over time (i.e., the specific target camera for that frame is set as the origin with an identity rotation).

LaCT-NVS [66] and NSTM (Ours): Stateful models must maintain coherent camera coordinate system across time

and adapt to time-varying target cameras. Therefore, for LaCT and NSTM, we relativize all cameras in the sequence to the *first* target camera ($t = 0$), ensuring a stable, global coordinate space for the fast-weights to operate within.

Token-Mem [49]: As explained in Sec. 4.1, *Token-Mem* is a custom novel view synthesis model whose memory mechanism is based on CUT3R [49]. As a stateful method, it also requires a temporally consistent coordinate frame, and we use the same coordinate system as described above for LaCT and our NTSM. Architecture-wise, we follow the original CUT3R [49] implementation to use a state size of 768 tokens each with 768 feature channel. Differing from [49], we optimize L2 and LPIPS losses consistent with the other methods compared.

B.4. Evaluation Protocol Details

Evaluation Dataset Filtering. Across all our long-term evaluation protocols—including the Memory Stress Test and ablation study (in the main paper), the Memory from Natural Rotations Test (Sec. A.1), and the General Novel View Synthesis Test (Sec. A.2)—we stress-test model performance over extended time horizons (i.e., up to $t = 60$). Consequently, we filtered the originally provided MVHumanNet++ test set to strictly include sequences containing at least 60 frames. This curation resulted in a unified subset of 390 evaluation scenes, which differs from the standard MVHumanNet++ test split but is applied consistently across all our long-term evaluations.

Sequence Ordering for Natural Rotations. For the

Natural Rotations Test (Sec. A.1), we utilize 17 scenes featuring natural back-revealing motions. Because these rotations typically occur near the end of the raw videos, we temporally reverse the sequences. This ensures the unoccluded back is observed early, allowing us to rigorously assess long-term memory retention (up to $t = 60$) over the subsequent minute-long horizon

C. Online Inference Protocol

Decoupled Memorization and Synthesis Frequencies.

Unlike models that rigidly couple memory updates and queries at every timestep, NSTM operates on independent execution strides. For our evaluation, the annotated train and eval sequences provided by the MVHumanNet++ dataset are aggressively subsampled capturing only 1 frame every 25 frames from the raw 30 FPS videos (i.e., subsampled at 1.2 FPS). Consequently, the temporal gap between consecutive inputs is roughly 833 ms. We train and evaluate by memorizing every other frame within these sequences. Furthermore, MVHumanNet++ provides raw 30 FPS videos; for our supplemental results, we demonstrate NSTM’s real-time rendering capabilities by synthesizing 29 frames for every 1 step of memorization.

D. Detailed Model Architecture

Fig. 6 shows our NSTM layer in more details. During inference, we run the memorization step periodically while performing the synthesis step per-frame. Different from training time, there is no memory tokens and we extract the target tokens as output for all steps. More specifically, following LVSM [22], we channel-wise concatenate input RGB images with their corresponding per-pixel Plücker ray maps, using linear layers to patchify them into input tokens. Similarly, we patchify the Plücker rays of the target camera to form target tokens. These concatenated input and target tokens are then passed through a $L = 24$ Transformer blocks, each comprising a cross-view attention module, a TTT memory module, and a feed-forward MLP module. The cross-view attention module applies self-attention on the concatenated input and target tokens. Beyond a standard TTT layer, we introduce a learned weight α to balance the contribution from the *cross-view attention* module and the TTT memory module. Finally, we extract the processed target tokens and unpatchify them with an linear layer, applying a sigmoid activation to predict the target RGB images.

E. Foreground Matting and Background Augmentation

To facilitate clean isolation of the dynamic subject from the studio background, we output a 4-channel $RGB\alpha$ image. During training, we employ background augmenta-

tion: we composite both the ground-truth foreground and our model’s predicted $RGB\alpha$ output onto randomly sampled solid color backgrounds before computing the photometric loss. This augmentation forces the network to learn a highly accurate alpha matte. Consequently, at inference time, NSTM produces a clean, disentangled alpha channel, enabling seamless compositing of the synthesized human subject into arbitrary novel environments.